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Christian College of Engg. & Tech. BHILAI

328713 (28)

BE (7th Semester)

Examination, Nov.-Dec., 2013

Branch : EEE, Et & T

MICROCONTROLLER & EMBEDDED SYSTEM

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : All questions are compulsory including part a and attempt any two parts from rest. Assumption can be made suitably with proper justification.

Q. 1. (a) Draw pin diagram of 8051. 2

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(b) Explain microcomputer system using GP microprocessor and microcontroller. What are major differences ? State advantages using microcontrollers ? 7

(c) State major differences between microcontroller 89c51, 8751, and ds5000 ? 7

(d) Give the criteria for choosing perfect microcontroller. 7

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Q. 2. (a) Give the names of various registers of 8051 ? 2

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(b) Write an ALP to find out equivalent ASCII code of ten bytes of Hex nos Residing from location 30h onwards in inbuilt RAM ? Store the result from same address. 7

(c) Write an ALP to monitor port pin p2.2 continuously and if p2.2 is high then Send 0FFH to port p0. if p2.2 is low then send 00h to port p0 also send low to High signal on port pin p2.3 ? 7

(d) Explain bit addressable area and its function. 7

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Q. 3. (a) Mention types of interrupts are available in 8051. 2

(b) Write an ALP to generate time delay of 1 sec using timer0 if XTAL = 11.0592MHZ ? 7

(c) Program timer0 to be an event counter. Use mode 1 to display the binary count on port p0 and p2 continuously ? Let initial count = 20000. 7

(d) Write a program in which every 2 seconds, the LED connected to P2.7 is turned off and on four times, while at the same time the

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8051 is getting data from P1 and sending it to P0 continuously. The on/off delay should be 50msec.

Q. 4. (a) What are RS232 & RS422 standards. 2

(b) How can we interface external memory larger than 64k with 8051 ? Explain with proper example and interfacing diagram ? 7

(c) Interface 8K DATA ROM and 4K DATA RAM with 8051. Write and ALP to access 10 bytes from external DATA ROM, bytes are residing from location 2000H. 7

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(d) What do you mean by on chip and off chip ROM ? How can program code accessed in 8031 ? Explain with interfacing diagram of 8031 and off chip ROM ? 7

Q. 5. (a) State different addressing modes of 8096. 2

(b) Explain with proper diagram, the interfacing between 8051 and ADC 0804 ? 7

(c) What do you mean by embedded system ? Explain embedded cards. 7

(d) Perform following calculations for interfacing of stepper motor 7

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(i) If motor takes 90 steps to make one revolution, what is step angle ?

(ii) Calculate the number of steps per revolutions for step angle of 7.5 degrees.

(iii) If step angle is known how can we set speed RPM of stepper motor ?

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